

Functional Analysis — Homework 4

Connor Mooneyhan

Problem 1. Fix $1 \leq p \leq \infty$ and let $S_L : \ell^p \rightarrow \ell^p$ be the left-shift operator:

$$S_L(x_1, x_2, x_3, \dots) = S(x_2, x_3, x_4, \dots).$$

Show that S_L is bounded and compute its operator norm. Does the equation $S_L(x) = x$ have any solutions other than $x = 0$?

Problem 2. Show that the closed unit ball of c_0 , which is the set

$$B_1[0] = \{x \in c_0 : \|x\|_\infty \leq 1\},$$

is a closed convex set with no extreme points.

Problem 3. Recall that $C^0[0, 1]$ is the space of continuous functions $f : [0, 1] \rightarrow \mathbb{R}$ equipped with the sup norm

$$\|f\|_\infty = \sup_{x \in [0, 1]} |f(x)|.$$

Find the extreme points of the closed unit ball $B_1[0]$ of this space. Conclude that $B_1[0]$ is **not** the closure of the convex hull of its extreme points (it does not satisfy the conclusion of Krein-Milman's theorem).

Problem 4. Let X be a vector space equipped with two norms, call them $\|\cdot\|_1$ and $\|\cdot\|_2$, such that X is Banach with respect to either of these two norms. Show that if these norms are *comparable*, i.e. if there exists a constant $C > 0$ such that

$$\|x\|_1 \leq C\|x\|_2, \quad \forall x \in X,$$

then they are *equivalent*, i.e. there also exists another constant $C' > 0$ such that

$$\|x\|_2 \leq C'\|x\|_1, \quad \forall x \in X.$$

Problem 5. Let X be a Banach space and $Y, Z \subset X$ be closed subspaces such that

(1) $Y \cap Z = \{0\}$.

(2) $Y + Z = X$ (every $x \in X$ can be written as $x = y + z$ with $y \in Y, z \in Z$).

Show that X is isomorphic to $Y \oplus Z$. In this situation, we say that Y (and by symmetry also Z) is a **complemented** subspace of X .

Problem 6. Let X be a normed space. In the language of the previous problem, show that any *finite-dimensional* subspace $Y \subset X$ is complemented.